

EE3621: Digital Signal Processing

Lecture 6: Inverse Z-Transform & Stability Analysis (III B.Tech EEE)

Lecture Notes (40-Minute Plan)

1 Lecture Plan & Objectives (40 Minutes Breakdown)

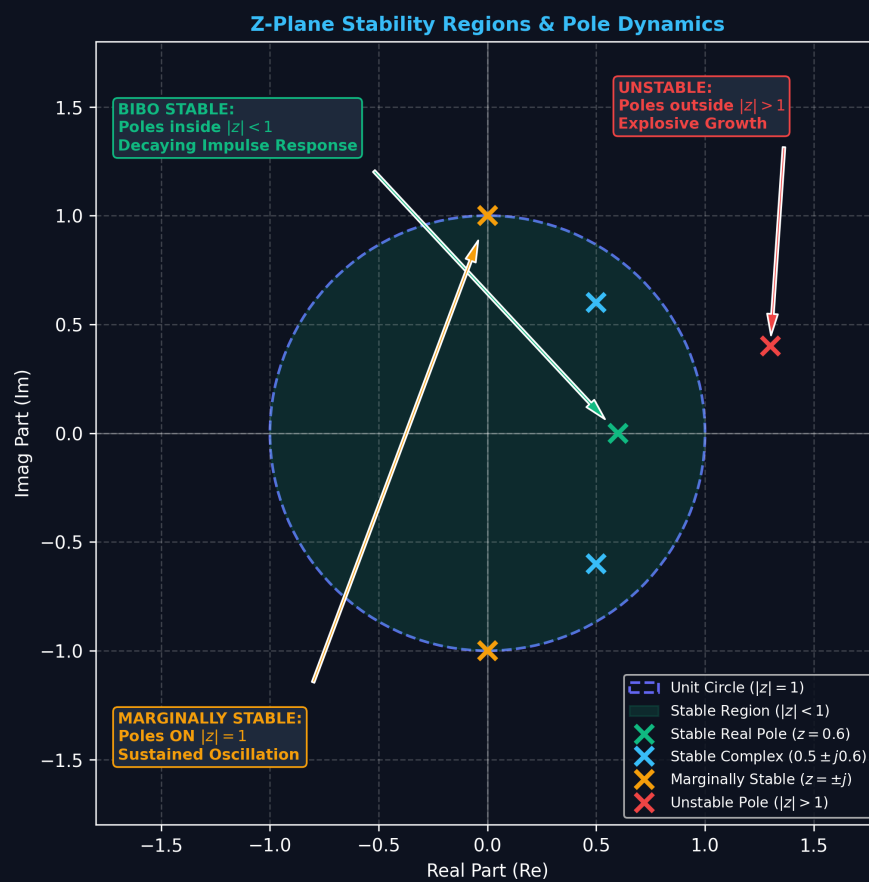


Figure 1: Z-Plane Stability Regions & Pole Dynamics: Stable ($|z| < 1$), Marginally Stable ($|z| = 1$), Unstable ($|z| > 1$).

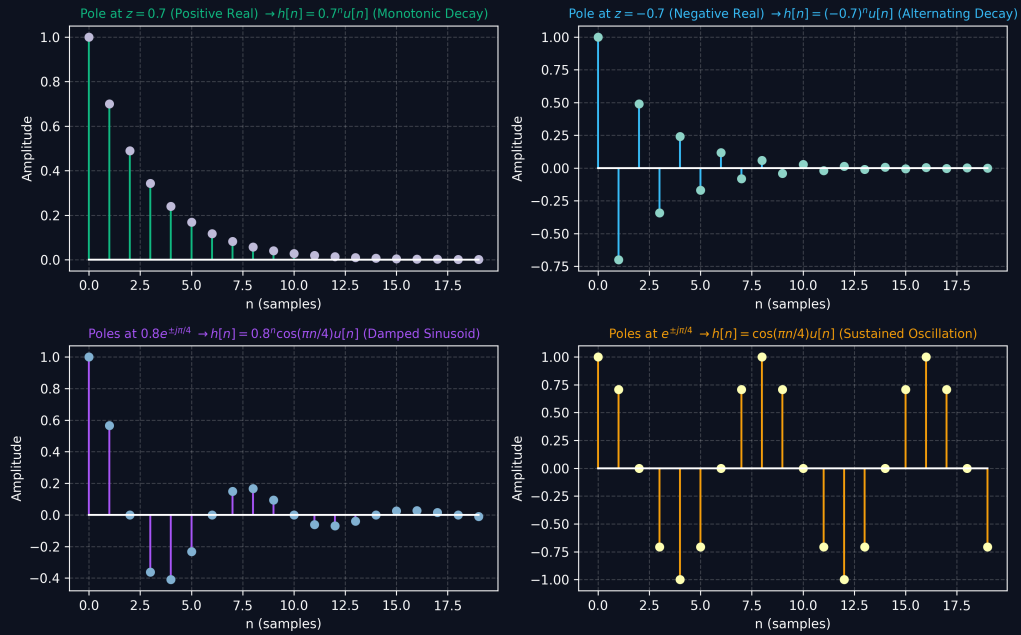


Figure 2: Time-Domain Impulse Response Modes corresponding to Pole Locations in the Complex Z-Plane.

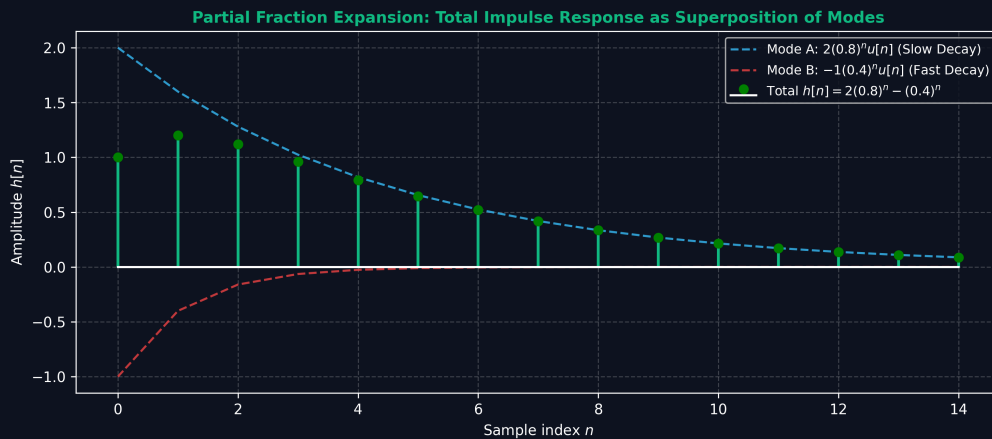


Figure 3: Partial Fraction Expansion: Synthesizing the Complete Impulse Response as a Superposition of Modes.

- **00:00 – 05:00 (5 mins):** Motivation: Why do we need the Inverse Z-transform? Uniqueness of ROC.
- **05:00 – 18:00 (13 mins):** Method 1: Partial Fraction Expansion (Distinct and Repeated Poles).
- **18:00 – 25:00 (7 mins):** Method 2: Power Series Expansion (Long Division for causal vs. anti-causal).
- **25:00 – 30:00 (5 mins):** Method 3: Contour Integration (Residue Theorem).
- **30:00 – 38:00 (8 mins):** Stability & Causality Analysis of LTI Systems based on poles.
- **38:00 – 40:00 (2 mins):** Checkpoint & Summary.

2 Overview of the Inverse Z-Transform

The Inverse Z-transform maps $X(z)$ and its ROC back to a discrete sequence $x[n]$:

$$x[n] = \mathcal{Z}^{-1}\{X(z)\} \quad (1)$$

The algebraic expression $X(z) = \frac{z}{z-a}$ corresponds to:

- $x[n] = a^n u[n]$ if ROC is $|z| > |a|$.
- $x[n] = -a^n u[-n-1]$ if ROC is $|z| < |a|$.

Specification of the ROC is critical to ensure uniqueness.

3 Method 1: Partial Fraction Expansion

3.1 Distinct Poles

For distinct poles p_k :

$$\frac{X(z)}{z} = \sum_{k=1}^N \frac{A_k}{z - p_k} \quad \text{where} \quad A_k = \left[(z - p_k) \frac{X(z)}{z} \right]_{z=p_k} \quad (2)$$

3.2 Repeated Poles

For a pole p_1 repeated m times:

$$\frac{X(z)}{z} = \sum_{k=1}^m \frac{A_{1k}}{(z - p_1)^k} + \sum_{i=2}^N \frac{A_i}{z - p_i} \quad (3)$$

where residues are calculated using derivatives:

$$A_{1k} = \frac{1}{(m-k)!} \left[\frac{d^{m-k}}{dz^{m-k}} \left((z - p_1)^m \frac{X(z)}{z} \right) \right]_{z=p_1} \quad (4)$$

4 Method 2: Power Series Expansion (Long Division)

We divide the numerator by the denominator:

$$X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n} \quad (5)$$

- **Causal** ($|z| > |p|$): Divide in descending powers of z (e.g. $1 + c_1 z^{-1} + c_2 z^{-2} + \dots$).
- **Anti-causal** ($|z| < |p|$): Divide in ascending powers of z (e.g. $c_{-1} z^1 + c_{-2} z^2 + \dots$).

5 Method 3: Contour Integration (Residue Theorem)

The formal inverse transform is evaluated as a contour integral:

$$x[n] = \frac{1}{2\pi j} \oint_C X(z) z^{n-1} dz = \sum \text{Residues of } [X(z) z^{n-1}] \text{ inside } C \quad (6)$$

For a pole $z = p$ of order m , the residue is:

$$\text{Res} \{ X(z) z^{n-1} \}_{z=p} = \frac{1}{(m-1)!} \lim_{z \rightarrow p} \frac{d^{m-1}}{dz^{m-1}} [(z-p)^m X(z) z^{n-1}] \quad (7)$$

6 Causality, Stability & Pole Locations

- **Causality:** The ROC of a causal LTI system transfer function $H(z)$ must be the exterior of a circle: $|z| > p_{max}$.
- **Stability:** The ROC of $H(z)$ must contain the unit circle ($|z| = 1$).
- **Causal & Stable:** All poles must lie strictly inside the unit circle ($|p_k| < 1$).

Stability boundaries and impulse response modes are illustrated in Figures 1 and 2.